

Mathematical Review: National Channel Territory Design and Staffing

An Analytical Assessment of the Optimization Model, Descaling Invariance, and Axiomatic Fairness Under Feasible Envy-Freeness

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September 13, 2026

Abstract

This review presents an exhaustive mathematical and axiomatic appraisal of the multi-channel sales territory design and wholesaler staffing optimization model documented in `docs/MODEL_FULL.md`. We analyze the hierarchical decomposition spanning state-level mixed-integer linear programming (Level 0), semi-discrete optimal transport with discrete graph repair heuristics (Level 2), and Nash social welfare bipartite matching (Stage 2). We establish the mathematical foundation of data descaling and prove why multiplicative opportunity normalization leaves the optimization argmax invariant. Furthermore, we investigate the fundamental tension in Nash bargaining at a zero versus non-zero disagreement point: while $d = 0$ historically anchored the classical Envy-Free up to One Item (EF1) guarantee of Caragiannis et al. (2019), that guarantee is already invalidated by hard territory capacity bands and spatial contiguity. We propose dropping symmetric anonymity in favor of **Feasible Envy-Freeness (FEF_x)** under Generalized Assignment Constraints (Barman et al., 2023) paired with asymmetric claims-scaled reservation utilities, resolving the flat-curvature pathology of the logarithmic welfare function while strictly guaranteeing bargaining set feasibility and incumbent book continuity.

1 Introduction and Problem Formulation

The national territory design problem formulated in `docs/MODEL_FULL.md` addresses the simultaneous partitioning of commercial sales opportunity across the 48 contiguous United States and Washington, D.C. ($S = 49$) into k balanced sales districts of approximately \$1B opportunity each. Concurrently, it models the allocation of incumbent sales representatives across four fine channels:

$$C = \{N_{WH}, N_{FI}, WH, FI\}$$

admitting seven permissible channel bundles:

$$\mathfrak{B} = \{N, WH, FI, WH_PLUS, FI_PLUS, WHFI, WHFI_PLUS\}$$

Formulated as a monolithic joint problem over 6,459 discrete ZIP codes and multiple business channels, the problem constitutes an intractable Mixed-Integer Non-Convex Non-Linear Program (MINLP). Topological contiguity constraints on planar graphs, multi-channel product-form allocations, and non-linear Nash bargaining welfare ($\max \prod g_i$) combine to preclude direct global optimization.

To render the computation tractable, the architecture establishes a **three-level hierarchical decomposition**:

1. **Level 0 (Macro Planning)**: A state-by-channel MILP solved lexicographically pass-by-pass on the state rook graph $G_S = (S, E_S)$.

2. **Level 2 (Micro Realization):** A transportation LP (power diagram) per split state, followed by topological repair heuristics on the ZIP-level Voronoi cell graph G_{cell} .
3. **Stage 2 (Wholesaler Staffing):** A maximum-weight bipartite matching on logarithmic gains solved via the Kuhn-Munkres (Hungarian) algorithm.

2 Detailed Mathematical Architecture

2.1 Level 0: Macro-Level Channel Planning (State $\times$ Channel MILP)

Level 0 aggregates opportunity mass $M_{s,c}$ to the state grain. For each slot $j \in J$ belonging to bundle B_j , the decision space comprises:

- $y_{s,j} \in [0, 1]$: Continuous share of state s allocated to slot j .
- $z_{s,j} \in \{0, 1\}$: Binary contact indicator ($y_{s,j} > 0$).
- $u_j \in \{0, 1\}$: Binary slot utilization indicator.
- $r_{s,j} \in \{0, 1\}$: Contiguity tree root indicator.
- $f_{a,j} \in [0, N - 1]$: Commodity flow on directed arc $a \in A_S$ where $N = |S| = 49$.

The core relations governing Level 0 are:

$$\text{Product-form Coverage: } \sum_{j:c \in B_j} y_{s,j} \leq 1 - p_{s,c} \quad \forall s \in S, c \in C \quad (1)$$

$$\text{Contact Sizing: } \eta z_{s,j} \leq y_{s,j} \leq z_{s,j} \leq u_j \quad \forall s \in S, j \in J \quad (2)$$

$$\text{Capacity Bands: } L_{B_j} u_j \leq \sum_{s \in S} W_{s,j} y_{s,j} \leq U_{B_j} u_j + \sum_{s \text{ capped}} a_s z_{s,j} \quad \forall j \in J \quad (3)$$

$$\text{Tree Contiguity: } \sum_{s \in S} r_{s,j} = u_j, \quad r_{s,j} \leq z_{s,j} \quad \forall j \in J \quad (4)$$

$$f_{a,j} \leq (N - 1) z_{\text{tail}(a),j}, \quad f_{a,j} \leq (N - 1) z_{\text{head}(a),j} \quad \forall a \in A_S, j \in J \quad (5)$$

$$z_{s,j} - N r_{s,j} \leq \sum_{a \in \delta^-(s)} f_{a,j} - \sum_{a \in \delta^+(s)} f_{a,j} \quad \forall s \in S, j \in J \quad (6)$$

where $W_{s,j} = \sum_{c \in B_j} M_{s,c}$, $p_{s,c}$ represents prior committed coverage, and $a_s = \max(0, W_s / \text{cap}_s - \tau_B)$ provides state-specific band-break allowances under split caps cap_s .

2.2 Level 2: Micro-Level Realization (Optimal Transport and Graph Repair)

Given state shares $y_{s,j}$, unsplit states are mapped entirely to district j . Split states are partitioned via the semi-discrete optimal transport LP:

$$\min_x \sum_{z \in s} \sum_j M_z (\|p_z - c_j\|^2 + \pi_{z,j}) x_{z,j} \quad (7)$$

$$\text{s.t. } \sum_j x_{z,j} = 1 \quad \forall z \in s, \quad \sum_{z \in s} M_z x_{z,j} = y_{s,j} M_s \quad \forall j \quad (8)$$

Centroids c_j are iteratively shifted via up to five Lloyd updates. Resulting disconnected components are reconnected post-LP via heuristic frontier expansion and bridging on the Voronoi dual graph G_{cell} with capacity slack $\sigma = 0.02\tau_B$.

2.3 Stage 2: Centralized Nash Social Welfare Staffing

Let A_j denote the spatial footprint of district j . The utility gain $g_{i,j}$ of representative $i \in R$ on district $j \in J_{\text{used}}$ is:

$$g_{i,j} = \sum_{z \in A_j} \sum_{c \in B_j} [c_1 S_i(z, c) + c_2 (T_{z,c} - S_i(z, c)) + c_{\text{free}} S_{\text{free}}(z, c) + \lambda M_{z,c}] \quad (9)$$

where $c_1 = 1 - \lambda$, $c_2 = \theta(1 - \lambda)$, and $c_{\text{free}} \in \{c_2, c_1, \lambda\}$. The matching problem:

$$\max_w \sum_{i \in R} \sum_{j \in J_{\text{used}}} \log(g_{i,j}) w_{i,j} \quad \text{s.t.} \quad \sum_i w_{i,j} \leq 1, \sum_j w_{i,j} \leq 1, \sum_{i,j} w_{i,j} = \min(|R|, |J_{\text{used}}|) \quad (10)$$

is solved globally via the Hungarian algorithm on costs $-\log(g_{i,j})$.

3 Data Descaling and Scale Invariance

A fundamental property of the operational pipeline is that the numerical solvers do **not** operate on raw dollar amounts.

3.1 The Descaled Data Representation

As implemented in `td/instance.py`, the raw instance data undergoes a canonical descaling transformation governed by $\kappa > 0$, defined as the median positive ZIP opportunity across the footprint:

$$m_{\text{rel}}(z) = \frac{M_z}{\kappa}, \quad s_i(z) = \frac{S_i(z)}{M_z} \quad (11)$$

Upon loading, the pipeline reconstructs:

$$M(z) = m_{\text{rel}}(z) = \frac{\text{real } M_z}{\kappa}, \quad S_i(z) = s_i(z) \cdot m_{\text{rel}}(z) = \frac{\text{real } S_i(z)}{\kappa} \quad (12)$$

Under this transformation, the national opportunity sum across CONUS is $W_0 \approx 3,268.41$ descaled units, and district target capacity becomes $\tau \approx 180\text{--}200$ descaled units (rather than $\approx \$10^9$ raw dollars).

3.2 Proposition: Exact Scale Invariance of Nash Welfare

Proposition 1 (Scale Invariance of Nash Social Welfare). *Let $\mathcal{I} = (G, (S_i)_{i=1}^n, S_{\text{free}}, M)$ be an instance. Scaling every currency field uniformly by $\kappa > 0$ yields instance $\kappa\mathcal{I}$. For every district assignment w and any perimeter penalty weight $\rho \geq 0$:*

1. $u_i^{\kappa\mathcal{I}}(z) = \kappa u_i^{\mathcal{I}}(z)$ and $g_{i,j}^{\kappa\mathcal{I}} = \kappa g_{i,j}^{\mathcal{I}}$.
2. $\sum_{i=1}^n \log g_{i,\pi(i)}^{\kappa\mathcal{I}} = \sum_{i=1}^n \log g_{i,\pi(i)}^{\mathcal{I}} + n \log \kappa$.
3. *The additive shift $n \log \kappa$ is strictly independent of the allocation w and assignment π . Consequently:*

$$\arg \max_{w,\pi} \left(\sum_{i=1}^n \log g_{i,\pi(i)}^{\kappa\mathcal{I}} - \rho P(w) \right) \equiv \arg \max_{w,\pi} \left(\sum_{i=1}^n \log g_{i,\pi(i)}^{\mathcal{I}} - \rho P(w) \right)$$

All objective differences, certificates, and optimality gaps are identical across $\mathcal{I}$ and $\kappa\mathcal{I}$.

3.3 Why Multiplicative Normalization Fails to Restore Curvature

A natural question is whether normalizing the utility function by overall opportunity (either instance-wide M_{tot} or district-level M_j) solves the flat-curvature issue of Nash welfare. Mathematically, it cannot:

1. **Global Normalization** ($g_{i,j}/M_{\text{tot}}$): By Proposition 1, dividing by a global constant simply subtracts $n \log M_{\text{tot}}$ from the objective. The argmax, gradients, and Hessian structure are completely untouched.
2. **District Normalization** ($g_{i,j}/M_j$): Expanding $g_{i,j}$:

$$\frac{g_{i,j}}{M_j} = \underbrace{(c_1 - c_2) \frac{S_i(V_j)}{M_j}}_{\Delta_{i,j}/M_j} + \underbrace{\left[c_2 \frac{T(V_j)}{M_j} + c_{\text{free}} \frac{S_{\text{free}}(V_j)}{M_j} + \lambda \right]}_{G_{0,j}/M_j}$$

Dividing by M_j scales numerator and denominator proportionally. The critical ratio:

$$\frac{\Delta_{i,j}/M_j}{G_{0,j}/M_j} = \frac{\Delta_{i,j}}{G_{0,j}}$$

is strictly **dimensionless**. Because $G_{0,j} \gg \Delta_{i,j}$ (ambient district opportunity and cross-selling dwarf individual book differences), the Taylor expansion remains linear:

$$\log \left(\frac{G_{0,j}}{M_j} + \frac{\Delta_{i,j}}{M_j} \right) \approx \log \left(\frac{G_{0,j}}{M_j} \right) + \frac{\Delta_{i,j}}{G_{0,j}}$$

Multiplicative scaling cannot alter the dimensionless ratio Δ/G_0 . Only an **additive translation** ($g - d$) can shift the operating point of $\log(x)$ toward its steep curvature regime.

4 The Disagreement Point Dilemma: Classical EF1 vs. Feasible Envy-Freeness

4.1 The Zero Threat Point and the Caragiannis Guarantee

In classical cooperative bargaining theory (Nash, 1950), the Nash product is maximized over gains above a disagreement point $d = (d_1, \dots, d_n)$:

$$\max_{\pi} \prod_{i=1}^n (u_i(\pi_i) - d_i) \quad (13)$$

In `docs/MODEL_FULL.md` and `docs/math_note/math_note.tex`, the baseline is fixed at $d = \mathbf{0}$. The foundational justification rests on the landmark theorem of Caragiannis et al. (2019):

Theorem 1 (Caragiannis et al., 2019). *For unconstrained fair division of indivisible goods under additive valuations and a zero disagreement point $d = \mathbf{0}$, any allocation maximizing the Nash Social Welfare $\prod_{i=1}^n u_i(A_i)$ is simultaneously:*

1. **Pareto Optimal (PO)**, and
2. **Envy-Free up to One Item (EF1)**: For any pair of agents i, k , there exists an item $z \in A_k$ such that:

$$u_i(A_i) \geq u_i(A_k \setminus \{z\})$$

4.2 Why Shifting to $d > 0$ Voids Classical EF1

The proof of Theorem 1 relies directly on local exchange ratios: if agent i envies agent k by more than one good, shifting item $z \in A_k$ to A_i strictly increases the Nash product:

$$\left(1 + \frac{u_i(z)}{u_i(A_i)}\right) \left(1 - \frac{u_k(z)}{u_k(A_k)}\right) > 1 \quad (14)$$

When an additive disagreement vector $d > \mathbf{0}$ is introduced:

$$\left(1 + \frac{u_i(z)}{u_i(A_i) - d_i}\right) \left(1 - \frac{u_k(z)}{u_k(A_k) - d_k}\right) \quad (15)$$

If rep i is driven near their reservation utility ($u_i - d_i \rightarrow \epsilon$), the marginal ratio $\frac{u_i(z)}{\epsilon}$ diverges to infinity. The optimizer responds by aggressively transferring goods from unconstrained agents to agent i , even when agent i already possesses substantial bundle value. As established in the literature ledger (`docs/foundations/LIT_economic-theory.md`, row A4), subtracting a constant strictly voids the Caragiannis guarantee; no deterministic indivisible EF1 analogue exists for $d > 0$.

4.3 The Reality: Capacity Bands Already Invalidate Classical EF1

Crucially, `docs/foundations/DOMAIN_economic-theory.md` (§2.1, §2.8) proves that the territory design problem **already violates the hypotheses of Theorem 1**:

1. **Quantity Capacity Bands:** Districts must satisfy $L \leq M(V_j) \leq U$. The optimizer cannot execute the arbitrary item transfers required by the Caragiannis exchange step without violating the capacity band.
2. **Topological Contiguity:** Items (ZIPs) cannot be transferred if doing so disconnects the territory graph.

Empirical measurement on the CONUS footprint demonstrates a representative utility spread of over 60% (`MODEL_U8-band`). Thus, classical unconstrained EF1 is already defunct on the ground.

4.4 Transition to Feasible Envy-Freeness (FEF_x)

Under Generalized Assignment Constraints, Barman et al. (2023) establish the appropriate axiomatic fairness concept:

Definition 1 (Feasible Envy-Freeness up to Any Item (FEF_x)). Let $\mathcal{F}_i$ denote the set of all feasible territories for representative i (satisfying drivability from home base, spatial contiguity, and capacity bands $[L, U]$). An allocation A is **Feasible Envy-Free up to Any Item (FEF_x)** if for every pair of agents i, k :

$$A_k \in \mathcal{F}_i \implies \forall z \in A_k, u_i(A_i) \geq u_i(A_k \setminus \{z\}) \quad (16)$$

If territory A_k is not feasible for representative i ($A_k \notin \mathcal{F}_i$), agent i cannot claim legitimate envy against A_k .

As succinctly formulated in the economics charter: “*A wholesaler cannot envy a bundle they could not work.*” Under FEF_x, cross-regional comparisons (e.g., an Ohio representative envying a Florida territory) are pruned by operational feasibility.

4.5 Dropping Anonymity: Asymmetric Entitlements and Claims

In sales force restructuring, representatives are not anonymous participants with equal claims to unowned goods. They hold **asymmetric historical entitlements** defined by their legacy books $S_i(Z)$.

We formulate the staffing problem as an **asymmetric claims problem** (O’Neill, 1982; Thomson, 2003). The objective becomes:

$$\max_{\pi} \sum_{i \in R} \log(g_{i,\pi(i)} - d_i) \quad (17)$$

subject to the **Bargaining Set Non-Emptiness Guard**:

$$d_i = \alpha_i S_i(Z) + d_{\text{floor}} \quad (18)$$

where $d_{\text{floor}} = \gamma \min_j G_{0,j}$ ($\gamma \in [0.5, 0.8]$) ensures a common safety net, and $\alpha_i \in (0, 1)$ is determined via proportional claims bankruptcy scaling:

$$\alpha_i = \min \left(1.0, \frac{M(\text{Region})}{\sum_{k \in \text{overlap}} S_k(\text{Region})} \right) \cdot (1 - \epsilon) \quad (19)$$

This formulation achieves three vital properties:

1. **Guaranteed Feasibility:** $g_{i,j} - d_i > 0$ for all valid district assignments.
2. **Restored Curvature:** Denominators drop from ≈ 200 to $\approx 10\text{--}20$, amplifying marginal resistance to book starvation by over $100\times$.
3. **FEFx Compliance:** Envy is bounded with respect to operationally feasible assignments.

5 Synthesis of Pipeline Assessments

Component	Mathematical Model	Theoretical Strengths	Structural Vulnerabilities
Level 0 Plan	Multi-Commodity MILP	Strict Pareto lexicographic hierarchy; compact single-commodity flow trees.	Continuous state shares $y_{s,j}$ ignore discrete spatial clustering of accounts.
Level 2 Cut	Transportation LP	Guaranteed Euclidean compactness; solved via dual simplex.	Disconnects spatial components across non-convex state geometries.
Level 2 Repair	Discrete Graph Heuristics	Restores physical contiguity on G_{cell} .	Heuristic; introduces $\pm 2\%$ capacity band slack.
Stage 2 Staffing	Nash Welfare Matching	Scale-invariant; polynomial solve via Kuhn-Munkres.	At $d = 0$, flat curvature treats book disruption as near-zero cost.
Fairness Audit	FEFx Matrix Audit	Evaluates envy only over feasible territories $\mathcal{F}_i$.	Replaces classical unconstrained EF1, which is voided by capacity bands.

Table 1: Comprehensive Mathematical Assessment Matrix.

6 Recommendations for the Implementation Plan

1. **Asymmetric Claims-Centered Staffing (Refinement 1):** Replace the zero disagreement baseline in Stage 2 with the claims-scaled reservation floor $d_i = \alpha_i S_i + d_{\text{floor}}$, operating $\log(\tilde{g}_i)$ in its high-curvature regime.
2. **FEFx Fairness Audit Suite:** In `tools/measure/audits.py`, replace the unqualified 13×13 EF1 test with the three-way verdict matrix: plain EF1, FEFx with respect to capacity and contiguity bands, and proportionality $u_i(A_i) \geq u_i(Z)/k$.
3. **Terminal Stage 2 Re-Match Pass:** Implement a post-Level 2 matching pass on finalized ZIP sets A_j to eliminate the efficiency loss caused by continuous state-share approximations.
4. **Contiguity-Constrained Spatial Splitting:** Promote the graph-based ‘-split-cut contiguous’ partitioner on G_{cell} to eliminate post-hoc heuristic repair slacks.